STUMPRIDER · TRACEABILITY MATRIX

# StumpRider traceability matrix

Design Molecule Lab · designmolecule.com

## DOCUMENT CONTROL

| | | | |
|---|---|---|---|
| Document ID | SMR-RTM-001 | Version | 0.3 |
| Status | Draft | Date | 2026-10-25 |
| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/stumprider | | |
| Source | docs/03b-traceability.md @ 507b36d | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.3 | 2026-10-25 | Amish Chadha | Generated by .kit/trace.py |
| 0.2 | 2026-10-25 | Amish Chadha | Generated by .kit/trace.py |
| 0.1 | 2026-10-25 | Amish Chadha | Generated by .kit/trace.py |

# StumpRider traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 11 user requirements, 13 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

| User requirement | Product requirement | Title | Evidence | Verification | Status |
|---|---|---|---|---|---|
| UR-001 | PR-001 | Free a snagged net | SMR-CAL-001, SMR-BLD-001, SMR-STM-001, SMR-OPT-001 | Analysis (SMR-CAL-001 s4), TRL 3; staged-snag trials with the hook head beside the ring, TRL 4 | At risk: hooked 81 to 126 N, inside the pin [F1, F2]; wrapped 368 N to 2.3 kN [F3, F4], 162 to 493 N once worked back half a turn [F7, F8] |
| UR-002 | PR-002 | Reach | SMR-CAL-001, SMR-BLD-001 | Analysis (SMR-CAL-001 s5), TRL 3; measured at the trial site, TRL 4 | Met on paper: 4.18 m [R1]; tether 8 m [R3] |
| UR-003 | PR-003 | Limit the side pull | SMR-CAL-001, SMR-BLD-001, SMR-STM-001 | Analysis (SMR-CAL-001 s3), TRL 3; CalRig pull tests and heel tests by canoe class, TRL 4 | At risk: static heel 3.4 deg kneeling, 4.4 deg standing [H5, H6] on screening hydrostatics (5.1 deg with a 1.0 m beam [Q1]); 6.5 deg peak for a quick pull held just under the pin [H11], ruled out by the steady-lift rule and measured at TRL 4 (O6, decided 2026-10-25); measured righting-moment criterion added (O9, decided 2026-10-25) |
| UR-004 | PR-004 | Protect the net | SMR-CAL-001 | Mesh count after staged releases, TRL 4 | To verify at TRL 4: the pin caps the pull at 347 N [P3] |
| UR-005 | PR-005 | Fast to use | SMR-CAL-001 | Timed trials with fisher crews, TRL 4 | To verify at TRL 4 |
| UR-006 | PR-006 | Fit any net line | SMR-CAL-001, SMR-BLD-001 | Analysis and inspection (SMR-CAL-001 s5), TRL 3; fit check on sample lines, TRL 4 | Met on paper: hinged ring, 100 mm inside [R4] |
| UR-007 | PR-007 | Stows in a canoe | SMR-CAL-001, SMR-BLD-001, SMR-STM-001 | Analysis (SMR-CAL-001 s8), TRL 3; weighed at TRL 4 | At risk: met on paper on the bamboo local variant, 3.95 kg [V3] (decision 2A, 2026-10-04), but the 0.05 kg margin after F1 is inside the uncertainty (wet bamboo, zinc, paint; independent review); aluminium prototype 4.34 kg [M11] recorded, not a failure. Packed 1.55 m [M9] |
| UR-008 | PR-008 | Survive wet storage | SMR-CAL-001, SMR- | Analysis (SMR-STM-001 S11 to S13), TRL 3; field | Met on paper: galvanised steel, 6063 aluminium, stainless pins and rivets, polyester, polyethylene; treated bamboo |

| User requirement | Product requirement | Title | Evidence | Verification | Status |
|---|---|---|---|---|---|
| | | | BLD-001, SMR-STM-001 | durability check with partner crews, TRL 4 | and zinc-painted ferrules in the local variant to watch |
| UR-009 | PR-009 | Locally buildable | SMR-CAL-001 | Analysis (SMR-CAL-001 s9), TRL 3; build trial with a local smith, TRL 4 | At risk: about 8.1 h [T1], galvanising sent out |
| UR-010 | PR-010 | Affordable | SMR-CAL-001 | Costed BOM (SMR-CAL-001 s9, s10), TRL 3; local prices, TRL 4 | Not met: USD 66.90 bamboo local variant [V8] (USD 54.90 before O8); USD 95.50 aluminium prototype [V9] |
| UR-003 | PR-011 | Calibrated pin wire | SMR-CAL-001, SMR-BLD-001, SMR-STM-001 | Inspection of the reel and CalRig breaks (SMR-BLD-001 first checks), TRL 4 | To verify at TRL 4 (SMR-STM-001 S1, S16) |
| UR-003 | PR-012 | The pin is the weakest link | SMR-CAL-001, SMR-STM-001 | Analysis (SMR-CAL-001 s6), TRL 3 | At risk: aluminium prototype met on paper (smallest factor 2.21, pole buckling [S2]; hook head 3.1 after F1 [S12]; pole side bending 2.4 [S17]); hook head with the bight on the tip leg 1.58 at the bend [S13] and 1.87 in the straight shank [S13a] (O10: the approved gusset was stopped on 2026-10-25 because it does not reach the shank, back with Amish); bamboo pole buckling 1.49 [V5] |
| UR-011 | PR-013 | Marked and handed out with protection work | SMR-CAL-001, SMR-BLD-001 | Inspection (BOM line 19; SMR-BLD-001 safety stops), TRL 3; partner distribution plan, TRL 4 | Met by design for the rating mark; tag artwork and distribution plan to do with the partner |

## Gaps

None.